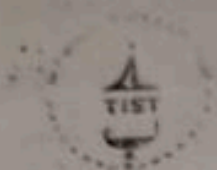


Date: 29/08/23

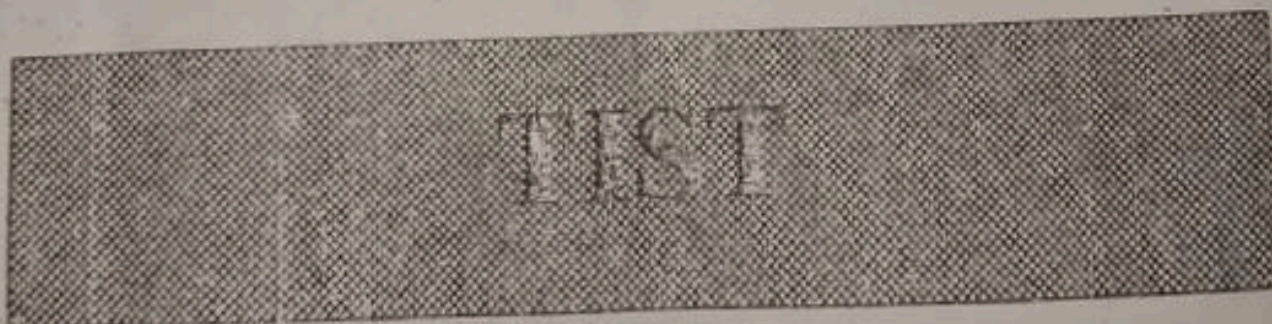


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INTERNAL ASSESSMENT MAIN ANSWER BOOKLET

	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	XIII
a	2			0	0	4	1	1					
b						0	4						
c													
d													
e													
f													
g													
h													
Total													
Grand Total	2					4	5	1					
TOTAL												12	



Name	Abhinand Dilip												
Branch & Semester	CSE-A 55												
Subject with Code	CST-307 Microprocessor and microcontroller												
Roll No.	4	University Reg. No:	T	0	C	2	3	C	5	0	01		
Signature of Invigilator	[Signature]						No. of Additional Sheets						

```

2) CLR
   MOV DI 2000H
   MOV SI 3000H
   MOV AX [SI]
   INC SI
   INC SI
   MOV BX [SI]
   MOV CL
   ADD AX, BX
   JNL 1018
   INC CL
   MOV DI [AX]
   INC DI
   INC DI
   MOV DI (CL)
   HLT

```

6) a)

1) b) The different types of addressing modes in 8086 are

- 1) Indexed addressing mode
- 2) Register addressing mode
- 3) Register indirect addressing mode
- 4) Base addressing mode
- 5) Direct addressing mode
- 6) Immediate addressing mode.

Give explanation with examples.

6) a) 8086

8086 is a 16 bit microprocessor consisting of ²⁰ bit datalines and it has a flags and pipelining is also possible, and is a general purpose based microprocessor.

It consist of mainly two units

a) Internal Bus unit consisting of

- CS - code segment - consisting of codes of samples
- DS - data segment - pointer holding data addresses
- SS - Stack segment - pointer controlling stack operations
- ES - Extra segment
- PI - pointer

b) Execution unit consisting of:

- Registers like

AX, BX, CX and DX to store and perform specific functions.

- It also contains the most crucial part of

a system i.e. ALU. which performs the logical

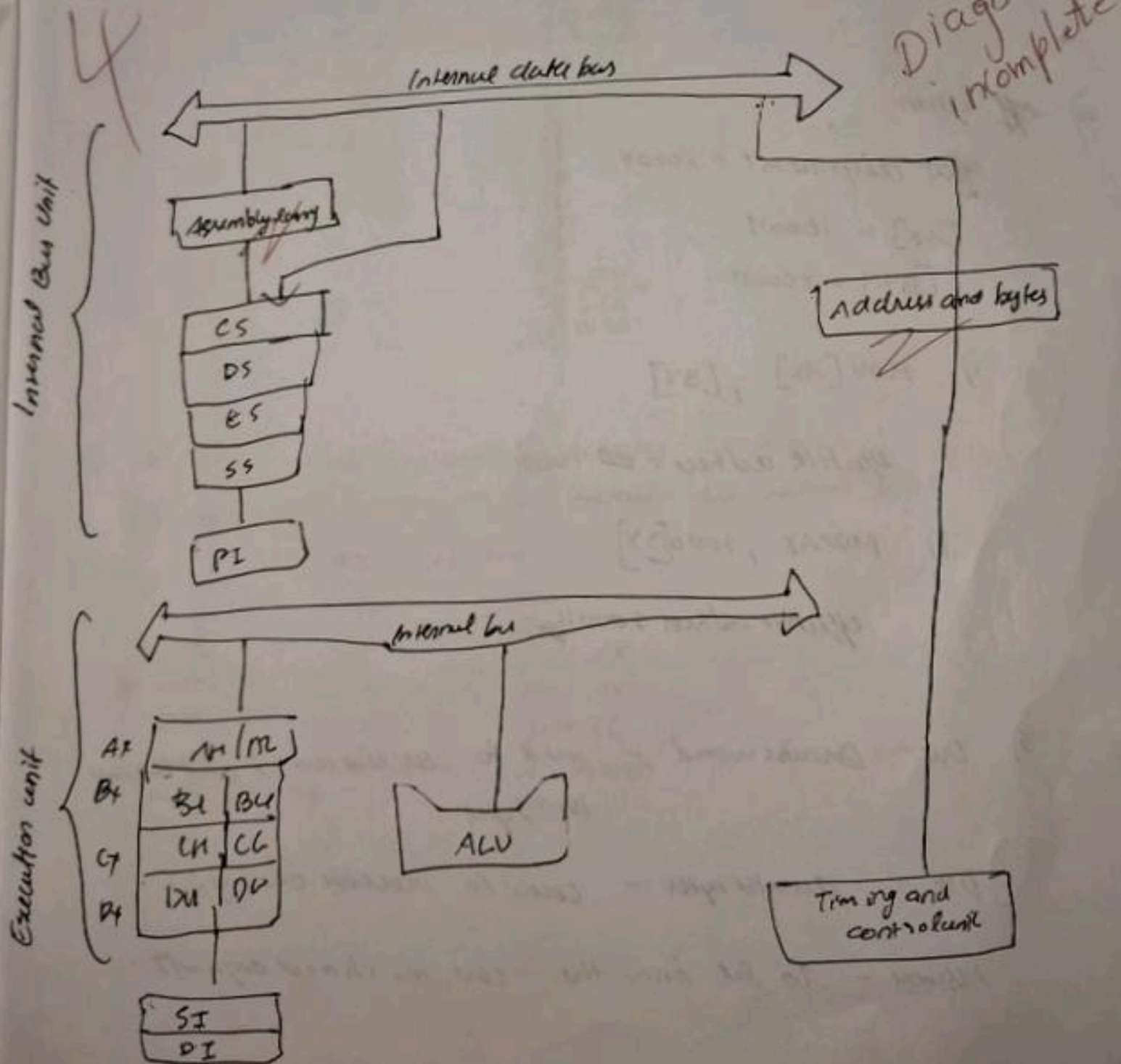
tasks of the computer.

- It also contains timing level control unit.

There are 9 types of flags in 80486 too.

- Zero flag
- Auxiliary carry flag
- carry flag
- overflow
- Parity flags etc.

Diagram:



1) 8085 Microprocessor:

- It is a 8 bit microprocessor
- It is an accumulator based microprocessor
- It has 16 address lines
- Pipelining is not possible
- It only has 64K memory
- It only has 5 flags

2) ~~if~~ given

effective displacement = 5000H

[AX] = 1000H

[BX] = 2000H

i) MOV [AX], [BX]

effective address = 2000H

ii) MOV AX, 5000[BX]

effective address 5000H

3) DW - Describe word - used to describe word and takes two bytes

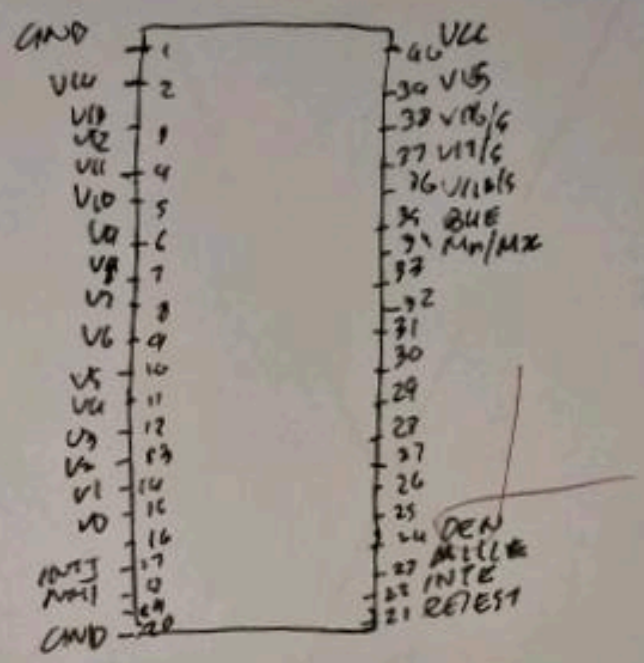
DB - describe bytes - used to reserve one byte.

Assume - To let know the CPU the shared segment.

- PROC - to start a procedure
- ENDP - to end a procedure
- ENDS - to end a segment
- EQ - to equals constants to variables
- SS - used to transfer and manipulate data between stacks
- ES - used to transfer data to IO devices



6.b) 8086 signals:



- 7.a) b) MUL - used to multiply between two registers
- c) DIV - used to divide between two values from given registers

- | | |
|--|--|
| <ul style="list-style-type: none"> CIR MOV AX MOV BX MOV CX MOV DX MUL AX, BX HLT | <ul style="list-style-type: none"> CIR MOV AX MOV BX MOV CX DIV AX, BX HLT |
|--|--|